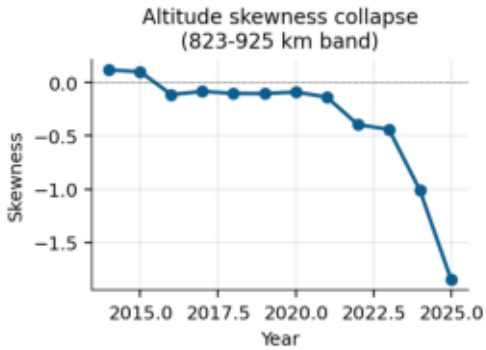
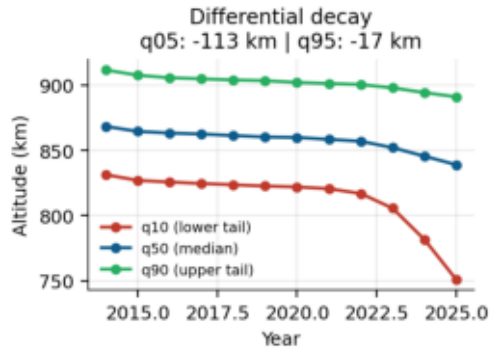
thermoLEO

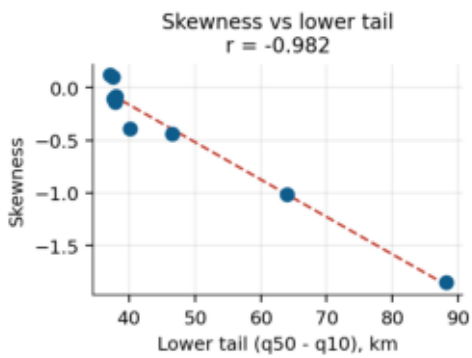
A TLE-based thermospheric density probe

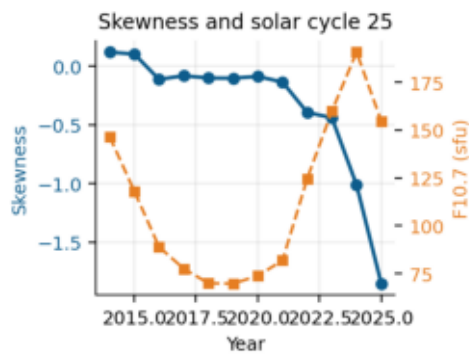
DOI: 10.5281/zenodo.23376365

github.com/Nat699/thermoLEO









PROBLEM

The upper thermosphere (800-1000 km) is poorly sampled by dedicated missions (CHAMP, GRACE, Swarm operate below 500 km). Empirical models rely on sparse assimilation and lack independent validation at these altitudes.

METHOD

We extract a density signal from the lower tail of the LEO satellite altitude distribution using only public TLE data. A fixed population of 4,108 Sun-synchronous satellites (1,369 in the 823-925 km band) is analyzed yearly from 2014 to 2025.

KEY RESULTS

- Skewness collapsed from +0.12 (2014) to -1.85 (2025), slope = -0.125/yr (p = 0.002)
- Lower tail (q05) decayed 6.6x faster than upper tail (q95): -113 km vs -17 km
- Correlation skewness x lower tail: r = -0.98 (bootstrap 95% CI [-0.999, -0.787])
- Multiple regression (skew ~ F10.7 + q10 + q50): R2 = 0.986

APPLICATIONS

- Global thermospheric density monitoring without dedicated instrumentation
- Satellite drag forecasting and reentry prediction (6-12 months ahead)
- Space weather nowcasting from public TLE archives (since 1957)
- Independent validation of JB2008 / NRLMSISE-00 at 800-1000 km

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